



STAFF SOFTWARE ENGINEER – FDE

Location: Chennai | Type: Full-time

About Newpage Solutions

Newpage Solutions is a global digital health innovation company helping people live longer, healthier lives. We partner with life sciences organisations which include, pharmaceutical, biotech and healthcare leaders, to build transformative AI and data driven technologies addressing real-world health challenges.

From strategy and research to UX design and agile development, we deliver and validate impactful solutions using lean, human-centered practices.

We are proud to be a '**Great Place to Work®**' certified company for the last three consecutive years. We also hold a top **Glassdoor rating** and are named among the "Top 50 Most Promising Healthcare Solution Providers" by CIOReview. As an organisation, we foster creativity, continuous learning and inclusivity, creating an environment where bold ideas thrive and make a measurable difference in people's lives.

Your Mission

We are looking for Staff Software Engineer who will Design and build scalable full-stack software systems that deliver measurable business value. Work directly with teams across Commercial, Manufacturing, and R&D to discover problems, navigate ambiguity, and ship practical solutions quickly. Develop AI-powered applications, including production RAG systems, LLM integrations, and optimized retrieval pipelines. Identify repeatable patterns in your work and collaborate with platform teams to generalize solutions into reusable frameworks. Lead technical initiatives, mentor engineers, and turn complex business challenges into reliable, production-ready systems.

What You'll Do

Responsibilities

- **Business:** Immerse in operations until you think like an insider. Rapidly acquire domain expertise through direct observation, translate between





business and engineering seamlessly, and mentor engineers in your area on immersion. Influence senior stakeholders effectively, manage complex stakeholder landscapes with competing agendas, and build trust rapidly with new stakeholders.

- **Delivery:** Lead rapid delivery initiatives across teams in your area, coach on prototype-first approaches, and establish trust through consistent fast delivery. Build complete applications rapidly across any technology stack, select the right tools for each problem, and define clear criteria for prototype-to-production transitions.
- **Generative AI:** Architect RAG systems for complex use cases across teams, implement advanced techniques (hybrid search, reranking, query expansion), mentor engineers on RAG best practices, and establish RAG standards. Lead evaluation strategy across teams, establishing annotation guidelines, training human-calibrated LLM judges, and building evaluation pipelines that connect tracing to datasets to experiments.
- **People:** Build high-performing teams across your area, navigate complex interpersonal dynamics, foster psychological safety, and create environments where diverse perspectives are valued. Influence through communication at all levels – from frontline to executive. Handle difficult conversations skilfully and train engineers in your area on effective communication.
- **AI-Augmented Development:** Optimise AI tool usage across teams in your area, train engineers on AI-augmented and agentic engineering workflows, evaluate new AI development tools, and establish practices that balance AI speed with verification rigour.
- **Scale:** Design complex multi-component systems end-to-end, evaluate architectural options for large initiatives across teams, guide technical decisions for your area, and mentor engineers on architecture. Create debt reduction strategies across teams, influence roadmap decisions to include debt work, and teach engineers when to accept debt for speed versus when to invest in quality.
- **Documentation:** Define documentation standards across teams in your area, create documentation systems and templates, train engineers on spec-driven development, and ensure documentation quality across projects. Lead pattern generalization initiatives, defining criteria for when to generalize versus keep custom.
- **Reliability:** Define reliability standards across teams in your area, drive post-incident improvements systematically, design capacity planning processes, and mentor engineers on SRE practices.





- **Process:** Lead lean transformations across teams in your area, design flow-optimised processes, coach engineers on lean principles, and balance speed with sustainability. Establish metrics that drive improvement.

Role Behaviours

- **Own the Outcome:** Drive accountability culture focused on outcomes not deliverables. Own business relationships and impact metrics across your function. Make trade-offs between custom solutions and generalisable work. There is no "I must run this by X." Ensure verification rigour for AI-generated code.
- **Be Polymath Oriented:** Champion cross-disciplinary learning. Create holistic solutions spanning technical and business domains. Embody the Renaissance Engineer ideal. Translate specialised knowledge into accessible explanations. Think like a business insider.
- **Communicate with Precision:** Create spec-driven development practices. Mentor others on precise communication. Span C-level executives to frontline workers. Drive clarity as a core value across your function. Represent the organisation externally.
- **Don't Lose Your Curiosity:** Drive team curiosity through challenging questions. Create environments where exploration and experimentation are encouraged. Model problem discovery orientation. Seek out ambiguity rather than avoiding it.
- **Think in Systems:** Shape systems design practices across your function. Conduct chaos engineering experiments. Influence cross-team architecture decisions. Create clarity from complexity. Bridge technical systems with business processes.

Practitioner-level Skills

- **Architecture & Design:** You design complex multi-component systems end-to-end, evaluate architectural options for large initiatives across teams, guide technical decisions for your area, and mentor engineers on architecture. You balance elegance with delivery needs.
- **Code Quality & Review:** You establish and enforce quality standards across teams in your area. You mentor engineers on effective code review, ensure verification depth for AI-assisted development, and drive testing strategies.
- **Full-Stack Development:** You build complete applications rapidly across any technology stack for teams in your area. You select the right tools for each problem, balance technical debt with delivery speed, and mentor engineers on full-stack development.



- **Problem Discovery:** You seek out undefined problems rather than avoiding them. You embed with users to discover latent needs, coach engineers in your area on problem discovery techniques, and turn ambiguity into clear problem statements.
- **Rapid Prototyping & Validation:** You lead rapid delivery initiatives across teams in your area, coach on prototype-first approaches, establish trust through consistent fast delivery, and define clear criteria for prototype-to-production transitions.
- **Retrieval Augmentation:** You architect RAG systems for complex use cases across teams, implement advanced techniques (hybrid search, reranking, query expansion), mentor engineers on RAG best practices, and establish RAG standards.
- **AI-Augmented Development:** You optimise AI tool usage across teams in your area, train engineers on AI-augmented workflows, evaluate new AI development tools, and establish practices that balance AI speed with verification rigour.
- **Multi-Audience Communication:** You influence through communication at all levels – from frontline to executive. You handle difficult conversations skilfully, train engineers in your area on effective communication, and represent teams across the function.
- **Business Immersion:** You immerse in operations until you think like an insider. You rapidly acquire domain expertise through direct observation, translate between business and engineering seamlessly, and mentor engineers in your area on immersion.
- **Multi-Audience Communication:** You influence through communication at all levels – from frontline to executive. You handle difficult conversations skillfully, train engineers in your area on effective communication, and represent teams across the function.
- **Stakeholder Management:** You influence senior stakeholders effectively across your area, manage complex stakeholder landscapes with competing agendas, build trust rapidly with new stakeholders, and shield teams from organisational friction.
- **Team Collaboration:** You build high-performing teams across your area, navigate complex interpersonal dynamics, foster psychological safety, and create environments where diverse perspectives are valued and heard.

Working-level Skills

- **DevOps & CI/CD:** You build complete CI/CD pipelines end-to-end, manage infrastructure as code, implement monitoring, and design deployment strategies for your services.



- **Cloud Platforms:** You design cloud-native solutions, manage infrastructure as code, implement security best practices, and make informed service selections. You troubleshoot cloud-specific issues.
- **AI Evaluation & Observability:** You design evaluation frameworks with custom evaluators tailored to your use case. You build golden datasets, establish annotation workflows with clear rubrics, and run experiments to compare prompt and model changes systematically.
- **Technical Debt Management:** You prioritise debt systematically based on risk and impact, balance debt reduction with feature work, and make pragmatic trade-offs. You know when to take on debt intentionally.
- **Data Integration:** You integrate multiple data sources independently, clean, messy datasets, handle inconsistent formats and missing values, and document data lineage. You troubleshoot integration failures.
- **Site Reliability Engineering:** You design observability strategies for your services, lead incident response, implement resilience testing, and conduct blameless post-mortems. You balance reliability investment with feature velocity.
- **Service Management:** You design service offerings with clear value propositions, manage service level agreements, improve service delivery based on user feedback, and communicate service status proactively.

Foundational-level Skills

- **AI Literacy:** You evaluate AI solutions critically for specific use cases. You understand bias, fairness, and hallucination risks. You make informed decisions about when AI helps versus when traditional approaches are better.
- **Data Modelling:** You create efficient data models that balance normalisation with query performance. You optimise queries, handle schema migrations safely, and choose appropriate storage technologies.
- **Technical Writing:** You create comprehensive documentation for complex systems. You write precise specifications that enable accurate AI-generated code, establish documentation practices for your projects, and ensure docs are discoverable.
- **Pattern Generalization:** You extract reusable components from field solutions, design appropriate abstractions that balance flexibility with simplicity, and collaborate with FDEs to validate generalised solutions in new contexts.
- **Knowledge Management:** You create searchable knowledge articles proactively, maintain team documentation, and organise information so others can find it.



- **Developer Experience:** You create friction logs documenting developer pain points. You write getting-started guides and contribute to improving existing golden paths based on user feedback.

What You Bring

- Bachelor's degree in computer science, Software Engineering, or related field with 7+ years of relevant professional experience.
- Deep production experience with Python and JavaScript/TypeScript across backend and frontend, with demonstrated ability to work comfortably across the full stack.
- Strong experience with modern frontend frameworks (such as Next.js or React) and backend API development.
- Extensive experience with cloud platforms (AWS preferred; Azure or GCP also valued), including infrastructure-as-code tools (such as CloudFormation or Terraform).
- Deep Working knowledge of multiple database paradigms, including relational databases (such as PostgreSQL), document databases, and key-value stores (such as Redis), with the ability to select the right storage technology for each problem.
- Strong experience with CI/CD pipelines (such as GitHub Actions), containerisation, and production deployment strategies.
- Demonstrable fluency with AI coding tools (such as Claude Code, Cursor, GitHub Copilot, or similar) and proven ability to design agentic engineering workflows and train teams on AI-augmented development practices.
- Hands-on experience architecting production generative AI applications (LLM integrations, vector databases, RAG systems, evaluation pipelines) is essential.
- Experience leading technical initiatives across multiple teams, mentoring engineers, and establishing engineering practices is required.
- Experience navigating ambiguous problem spaces, working directly with business stakeholders and end users, and shipping working solutions rapidly is required.
- Experience in an embedded, forward-deployed, or consulting-style engineering model is a strong plus.

Additional Notes



- Must have exceptional communication skills and demonstrated ability to influence at senior levels. This role requires comfort leading across ambiguous environments, building trust-based relationships with commercial stakeholders at all levels, developing junior and mid-level engineers, and making strategic decisions that balance delivery speed with architectural integrity. You will shape engineering practices across your area and be expected to lead multi-team initiatives.

What We Offer

At Newpage, we're building a company that works smart and grows with agility, where driven individuals come together to do work that matters. We offer:

- A people-first culture - Supportive peers, open communication and a strong sense of belonging
- Smart, purposeful collaboration - Work with talented colleagues to create technologies that solve meaningful business challenges
- Balance that lasts - We respect your time and support a healthy integration of work and life
- Room to grow - Opportunities for learning, leadership and career development, shaped around you
- Meaningful rewards - Competitive compensation that recognises both contribution and potential

Ready to Apply?

Let's build the future of health together. Apply below or reach out to:

shubhi.tiwari@newpage.io

